Kofi Mensah

Machine Learning Engineer

Contact

San Francisco, CA
kofi.mensah@example.com
(555) 010-0144

Skills

Languages
Python, SQL, C++, Bash

ML
PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, XGBoost

LLM
RAG, embeddings, vector databases (FAISS, Pinecone), LoRA fine-tuning, prompt evaluation

MLOps
MLflow, Kubeflow, SageMaker, Docker, Kubernetes, Airflow

Data
Spark, pandas, Snowflake, feature stores

Education

M.S. Computer Science (Machine Learning), Summit Ridge University (2019)

Certifications

AWS Certified Machine Learning - Specialty

Summary

Machine learning engineer who takes models from notebook to production. Recent work on retrieval-augmented generation and model serving. Interested in retrieval quality, evaluation and keeping models reliable after launch.

Experience

Machine Learning Engineer, *Quillmark AI* **Apr 2022 - Present**

- Built a RAG assistant over 80,000 internal support documents with FAISS and a fine-tuned embedding model; top-5 answer recall rose from 61% to 83% on a labeled set.
- Served transformer models with TorchServe on Kubernetes at 250 requests per second, with p95 latency of 90 ms.
- Created an evaluation harness that scores answers for groundedness and caught 3 regressions before release.
- Moved training jobs to SageMaker pipelines, cutting experiment turnaround from 2 days to 6 hours.
- Fine-tuned a 7B parameter open model with LoRA on 30,000 curated support conversations; human raters preferred it over the base model in 71% of comparisons.
- Reduced inference cost per query by 46% using dynamic batching, quantization to 8-bit and a response cache for repeated questions.
- Built monitoring for data drift and answer quality with Prometheus and Grafana, with alerts on groundedness scores.

Data Scientist, *Foxglove Analytics* **Jul 2019 - Mar 2022**

- Trained gradient boosted models for churn prediction that lifted retention campaign ROI by 22%.
- Built feature pipelines in Spark and Airflow processing 400 GB per day.
- Deployed a real-time fraud scoring model behind a FastAPI service with p95 latency of 35 ms.

PROJECTS

Paper reproduction: ColBERT retrieval *(PyTorch, FAISS)*

- Reproduced late-interaction retrieval results on MS MARCO and compared against BM25 and dense baselines.